PROJECT REPORT ON

Smart Home Ventilation System Using ESP32

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List of Principal Symbols and Acronyms

Acronym	Full Form
Blynk	Platform for IoT App Control
DHT11	Digital Humidity and Temperature Sensor
ESP32	Espressif Systems Microcontroller with Wi-Fi & Bluetooth
GPIO	General Purpose Input/Output
IoT	Internet of Things
PWM	Pulse Width Modulation
SG90	Servo Motor Model SG90
Wi-Fi	Wireless Fidelity
LCD	Liquid Crystal Display
PWM	Pulse Width Modultion
OLED	Organic Light Emitting Diode

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Abstract

In this project, a **Smart Home Ventilation System** has been designed and developed with the primary goal of maintaining **optimal indoor air quality** by intelligently managing humidity levels. At the core of the system lies the **ESP32 microcontroller**, a powerful and energy-efficient platform that enables both real-time processing and wireless communication. The system employs a **DHT11 sensor** to continuously monitor the ambient temperature and humidity within an indoor environment. Based on the sensed humidity data and a predefined threshold value, the ESP32 dynamically controls an **SG90 servo motor**, which mechanically opens or closes a **ventilation flap** to regulate air exchange within the room.

One of the key innovations in this system is the integration of **manual override functionality** using the **capacitive touch pin** of the ESP32. This allows users to directly control the state of the ventilation flap with a simple touch input, offering **flexibility and personalization** according to individual comfort or environmental changes. The system intelligently switches between automatic and manual modes, ensuring that user preferences are respected without compromising on automated climate control.

Furthermore, the system is designed to be **low-cost**, **compact**, and **modular**, making it highly suitable for a wide range of applications—from small apartments to office spaces. The use of **IoT-friendly features**, such as remote monitoring and potential future integration with platforms like **Blynk**, enhances its scalability and usability. Its energy-efficient design ensures long-term operation even in resource-constrained environments.

Overall, the Smart Home Ventilation System represents a **practical and innovative solution** for modern homes aiming to balance convenience, sustainability, and health. By combining environmental sensing, intelligent control, and user-friendly features, this project demonstrates how **affordable embedded systems and IoT technologies** can be harnessed to solve real-world problems in smart living environments.

Chapter 1

INTRODUCTION

1.1 Background :

As technology continues to advance, the concept of smart homes has emerged as a transformative trend in residential living. These homes are equipped with interconnected devices that automate various tasks, improve efficiency, and provide enhanced control over living environments. One of the most critical aspects of maintaining a comfortable living space is ensuring optimal air quality, which is directly influenced by factors 1 such as temperature and humidity. Inadequate ventilation leads to the accumulation of moisture in indoor environments, fostering the growth of mold and causing discomfort. It can also have long-term effects on health and the structural integrity of the building. Given these challenges, there is a growing need for intelligent systems that can automatically adjust ventilation based on realtime environmental conditions. By leveraging modern microcontroller-based automation and the Internet of Things (IoT), it is possible to create affordable, efficient, and easy-to-install solutions that can be integrated into any home or building. This project seeks to address these issues by developing a Smart Home Ventilation System that can autonomously monitor and control indoor ventilation based on humidity levels, ensuring a healthier and more comfortable living space.

1.2. Objective of the Project :

The primary goal of this project is to design a Smart Home Ventilation System that meets the following objectives: Automatic Humidity Control: The system will monitor real-time humidity levels using the DHT11 sensor and automatically adjust the ventilation by controlling a servo motor. This will open or close ventilation outlets to maintain optimal humidity levels. Manual Override: The system will include a feature to allow users to override the automatic control using the capacitive Touch pin on the ESP32 microcontroller. This provides flexibility for users to adjust ventilation based on personal preference or unique environmental conditions. Remote Monitoring and Control with Blynk: The system will support remote operation through the Blynk IoT platform, allowing users to switch between auto and manual modes, and view live humidity data from their smartphones. This adds convenience and enhances user control even from a distance.

Energy Efficiency and Cost-Effectiveness: The system will be designed to consume minimal power and utilize affordable components, making it a cost-effective solution for smart home environments.

Ease of Use: The system will be user-friendly, easy to install, and will not require complex configuration, making it accessible to homeowners with limited technical expertise.

1.3. Scope of the Project :

The Smart Home Ventilation System is mainly designed for use in small homes, apartments, or office rooms where maintaining good air quality and comfortable humidity levels is important. Many people face problems like high humidity or poor air circulation, especially in closed rooms. This system helps solve that by using automatic control based on real-time humidity. The project is low-cost, easy to use, and doesn't need any complex setup. It runs on its own using a humidity sensor and a servo motor to open or close a vent depending on the humidity level in the room. This makes it perfect for people who want to improve air flow without spending too much money or installing large systems. One of the key features is that it already supports remote monitoring and control using the Blynk app. Through a smartphone, users can view live humidity readings and switch between automatic and manual modes from anywhere, even if they are not at home. This adds convenience and control to the system. In the future, this system can be expanded by adding more sensors, such as temperature or gas sensors, and can be connected with other smart home devices. It can also be used in bigger spaces like classrooms, greenhouses, or small warehouses with just a few changes. 2 Overall, this project shows how simple electronics and IoT can be used together to build a useful, smart solution for everyday problems like ventilation and air quality.

1.4. Objective and Motivation :

Indoor air quality is often compromised in homes and commercial spaces due to inadequate or inefficient ventilation systems. Traditional ventilation methods require manual intervention, whether by adjusting windows or using fans, which is neither energy efficient nor reliable in maintaining consistent environmental conditions. Many existing systems that claim to be smart still rely on simplistic timers or basic sensors that do not account for dynamic environmental changes. There is a need for a truly intelligent system that can continuously monitor the indoor environment and autonomously control ventilation, ensuring optimal air quality with minimal user involvement.

1.5. Existing Systems :

While there are existing smart ventilation systems on the market, many of these solutions are expensive and often require complex installations or proprietary systems that are not compatible with open-source

hardware or platforms. Some systems use humidity sensors but still rely on manual control or rudimentary timers. In contrast, this project aims to bridge the gap by providing a low-cost, reliable, and fully automated solution that combines real-time monitoring with autonomous ventilation control. Additionally, the integration of user-friendly features, such as the capacitive Touch pin for manual override, sets this system apart from traditional solutions.

1.6. Scope of Present Work :

• Integration with IoT platforms like Blynk or Firebase for remote monitoring

• Real-time display via LCD or OLED screens

• Expansion to multi-zone (room-based) humidity sensing

• Incorporation of temperature-based control logic

• Voice assistant integration (e.g., Alexa or Google Assistant)

• Cloud-based logging and analytics dashboard

Chapter 2

COMPONENTS REQUIRED

2.1. ESP32 Dev Board

The ESP32 is a versatile and cost-effective microcontroller developed by Espressif Systems. It features dual-core processing, enabling it to handle more complex tasks simultaneously.

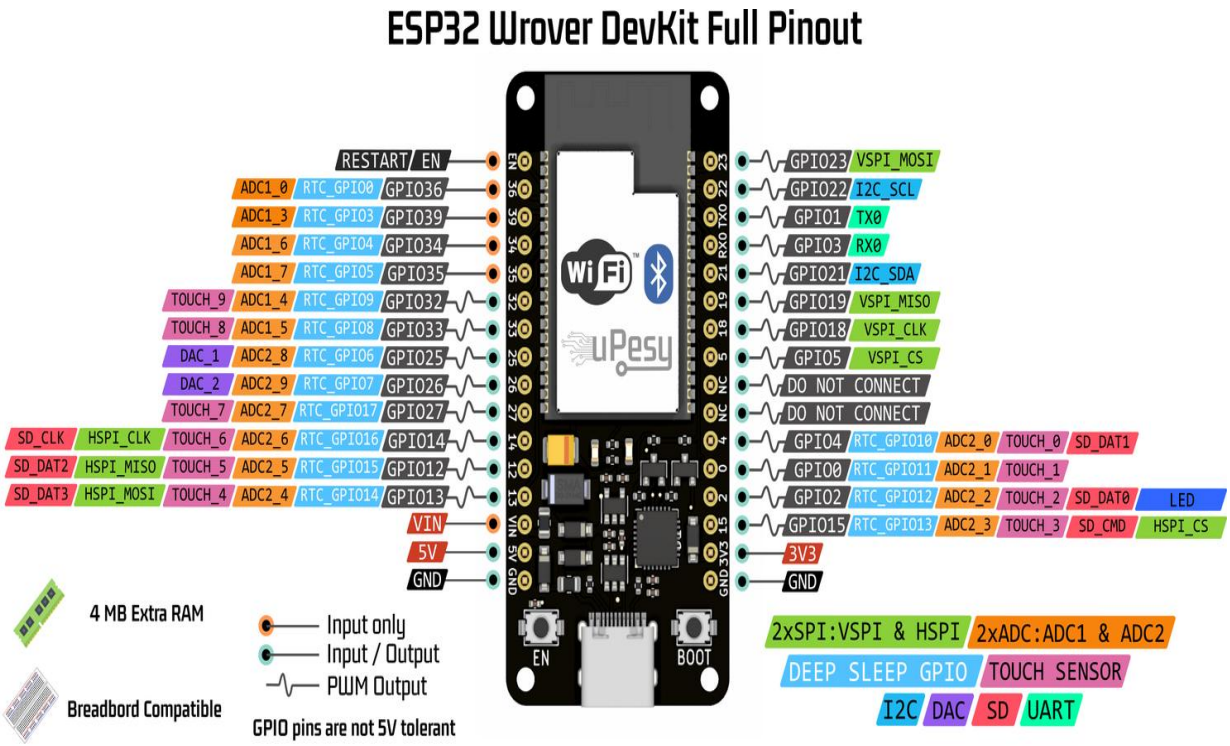

Figure 1

With integrated Wi-Fi and Bluetooth capabilities, the ESP32 is well-suited for a wide variety of IoT and automation applications. Its abundant digital and analog I/O pins offer flexibility for connecting sensors, actuators, and other peripherals. The microcontroller's low power consumption makes it an excellent choice for battery-operated devices or projects that require energy efficiency. Moreover, the ESP32 supports popular development platforms such as Arduino IDE and Micro Python, making it accessible to both beginners and advanced developers. This combination of performance, connectivity, and ease of use makes the ESP32 a popular choice for smart home systems, wearable devices, and sensor networks. Its robust performance

ensures real-time tasks are executed efficiently, while its extensive community support provides developers with numerous resources and libraries for rapid project development.

2.2. DHT11 Sensor:

The DHT11 is a popular, low-cost sensor used to measure temperature and humidity, ideal for IoT and home automation projects. It provides a simple digital output, making it easy to interface with microcontrollers like the ESP32. The sensor is designed for applications where precise measurements are not a primary concern but accurate monitoring of environmental conditions is still needed.

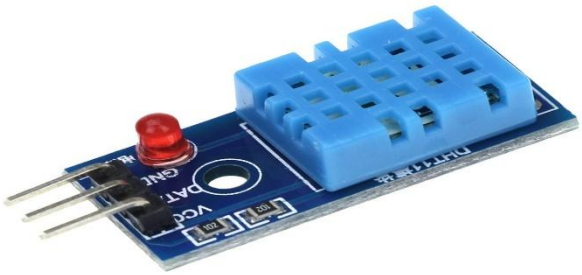

Figure 2

One of the main advantages of the DHT11 is its affordability and ease of use, which makes it a great option for beginners working with electronics and IoT projects. Its small size allows it to fit into compact devices, and its low power consumption makes it suitable for battery-powered systems. Although it has some limitations in terms of speed and accuracy compared to more advanced sensors like the DHT22, the DHT11 is still perfect for simple applications such as weather stations, smart home devices, or environmental monitoring systems. It is compatible with popular development platforms like Arduino IDE, making it accessible to a wide range of users, from hobbyists to professionals. Additionally, the DHT11 is capable of providing temperature readings within a range of 0-50°C and humidity readings suitable for monitoring conditions in a variety of environments. While it may not be ideal for highly precise applications, it offers reliable performance for basic environmental monitoring tasks. Its simplicity and affordability make it an excellent choice for projects requiring general temperature and humidity data, such as smart home systems or weather monitoring.

2.3. Servo Motor (SG90):

A servo motor is a type of electric motor used for precise control of angular position, velocity, and acceleration. It consists of a small motor, a gear train, and a control circuit that allows the motor to rotate to a specific angle based on input signals.

Figure 3

Servo motors are commonly used in applications that require high precision, such as robotics, automation systems, and control systems in IoT projects. In a typical servo motor, the input signal is a Pulse Width Modulation (PWM) signal, where the width of the pulse determines the position of the motor shaft. A shorter pulse moves the motor to one position, and a longer pulse moves it to another. The accuracy of servo motors makes them ideal for tasks that involve precise positioning, such as controlling the movement of robotic arms, steering mechanisms in drones, or adjusting the position of a camera. Servo motors are available in various sizes and torque ratings, allowing them to be used in both small-scale applications and heavy-duty systems. They are energy-efficient and have a relatively low power consumption when compared to other motors. With their reliable performance and ability to hold a position without consuming continuous power, servo motors are an essential component for many precision-based tasks in automation and control systems.

2.4. Breadboard and Jumper Wires:

Jumper wires and breadboards are essential tools in electronics for prototyping and testing circuits. A breadboard is a rectangular plastic board with a grid of holes, allowing for easy insertion and removal of electronic components without the need for soldering.

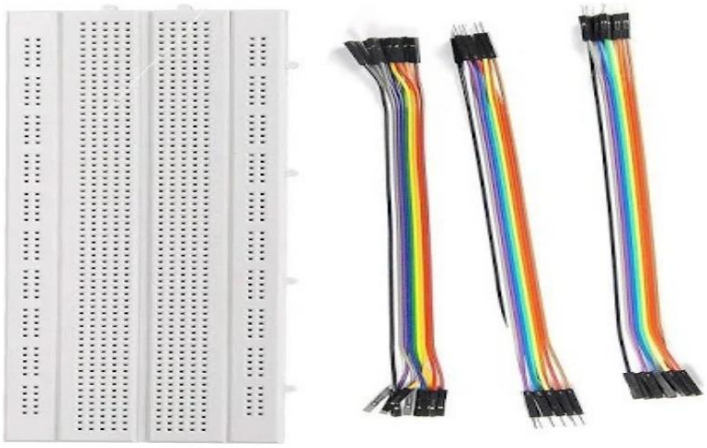

Figure 4

It provides a temporary platform to build and test circuits, making it ideal for experimentation and learning. The grid is divided into rows and columns that help organize the components and create stable connections between them. Jumper wires, which are flexible, insulated wires with connectors at both ends, are used to establish connections between components on the breadboard. They come in various lengths and types, including male and female connectors, which provide versatility in making temporary connections between circuit elements. The male connectors are typically inserted into the breadboard's rows, while the female connectors are used to connect to other components or microcontroller pins. These tools are crucial for electronics enthusiasts, allowing them to quickly build, modify, and test circuits without permanent soldering. Jumper wires are color-coded to aid in organization and identification of connections. Together, breadboards and jumper wires allow for quick prototyping, troubleshooting, and learning about circuit design in a flexible, non-permanent way.

Chapter 3

SYSTEM OVERVIEW

The proposed Smart Home Ventilation System is designed to automatically regulate indoor humidity levels using a compact, energy efficient microcontroller-based solution. The core of the system is built around the ESP32 microcontroller, which integrates real-time environmental sensing, decision-making logic, and actuation control. A DHT11 sensor is interfaced with the ESP32 to measure ambient temperature and relative humidity. These readings are continuously monitored, and when the humidity exceeds a predefined threshold, the ESP32 activates a servo motor to adjust a ventilation flap or window—thereby enhancing air circulation and improving indoor air quality.

To improve user control and flexibility, the system incorporates the ESP32's capacitive touch functionality as a manual override mechanism. This touch pin allows users to enable or disable the automated servo motor response based on specific requirements or personal preferences. When the override is triggered, the servo motor resets to a default position, suspending further automated actions until the override is deactivated.

Additionally, the system integrates the Blynk IoT platform, allowing users to remotely monitor and control the system through their smartphones. Via the Blynk app, users can view live humidity readings, switch between automatic and manual modes, and manage ventilation remotely from anywhere. This added functionality enhances user convenience and control, making the system even more adaptable to the user's needs.

The entire system operates without reliance on external servers or connectivity, making it a standalone and reliable solution for smart home environments. Its modular design also allows for scalability and integration with broader home automation systems. Overall, the system provides an intelligent and user-friendly approach to maintaining comfortable and healthy indoor environments.

3.1. Automatic Mode :

The Automatic Mode is the core functionality of the Smart Home Ventilation System, designed to operate autonomously based on real-time environmental conditions. In this mode, the ESP32 microcontroller continuously monitors the humidity levels in the indoor environment through the data provided by the DHT11 sensor. The system is programmed with a user-defined threshold for humidity, which can be adjusted based on the user's preference or environmental needs.

Once the humidity surpasses the set threshold (e.g., 85 Percent), indicating that the air is becoming too humid for optimal comfort or health, the system automatically activates the servo motor. The servo motor then drives a ventilation flap or a similar mechanical mechanism to open, allowing fresh air to circulate and reduce the humidity levels inside the space. This process facilitates better air quality and prevents the accumulation of moisture that could otherwise lead to mold growth or discomfort.

The system continually tracks the humidity levels, ensuring that the ventilation process is only triggered when necessary. When the humidity levels fall back within the acceptable range (e.g., below 70 percent), the ESP32 automatically sends a signal to close the ventilation flap. This ensures that the system does not waste energy or cause unnecessary air exchange once the room has returned to optimal conditions.

The Automatic Mode offers several key benefits:

Autonomous Operation: Once set up, the system requires no further user intervention, making it highly convenient and efficient.

Energy Efficiency: By only activating the ventilation when needed, the system optimizes energy consumption, ensuring that ventilation is used only when it will have the most benefit, reducing wear and tear on mechanical components.

Environmental Comfort: By maintaining consistent humidity levels, the system helps ensure a comfortable indoor environment, reducing the potential for excess moisture and the discomfort it can cause.

Health Benefits: Maintaining a balanced humidity level in a living space is essential for good health. The system helps prevent the negative effects of high humidity, such as mold growth, allergens, and respiratory issues.

This mode is particularly useful in spaces such as bedrooms, bathrooms, kitchens, and basements, where humidity levels can fluctuate frequently due to various activities like cooking, showering, or seasonal changes. The Smart Home Ventilation System in Automatic Mode ensures that the indoor environment remains consistently comfortable without requiring ongoing attention from the user. Additionally, the system's algorithm is designed to be adaptable, allowing users to adjust the threshold levels of humidity for different times of the day or specific conditions, such as seasonal variations. This ensures that the system operates optimally across various conditions while meeting the user's preferences.

By integrating Automatic Mode, this project aims to provide a seamless, efficient solution for maintaining ideal air quality and humidity levels in modern homes, contributing to both environmental sustainability and user comfort.

3.2. Manual Override Mode:

The Manual Override Mode is a crucial feature of the Smart Home Ventilation System, providing users with the flexibility to manually control the ventilation system, independent of the automatic sensor-based functions. This mode is particularly beneficial in scenarios where user intervention is necessary, such as during maintenance, extreme weather conditions, or personal preferences that require manual operation.

In this mode, a capacitive touch module, connected to GPIO27 of the ESP32 microcontroller, serves as the control interface. The capacitive touch module allows users to switch between the Automatic Mode and Manual Override Mode with a simple touch. The touch interface provides a seamless, intuitive way to toggle the system's operational mode, ensuring ease of use and quick access for the user.

When the Manual Override Mode is activated, the system no longer responds to sensor inputs from the DHT11 sensor. Instead, the user gains complete control over the servo motor's operation. The servo motor

toggles between predefined open and closed positions, irrespective of the ambient humidity levels detected by the sensor. This gives the user full autonomy to control the ventilation flap's position according to their specific needs or desires at that moment.

Key Benefits of the Manual Override Mode:

User Flexibility: The Manual Override Mode provides users with the ability to adjust the system's operation manually, giving them complete control over the ventilation process. This is particularly helpful in scenarios where sensor-driven automation might not be desirable.

Temporary Control for Specific Scenarios: There are instances when the automatic functionality might not be suitable or when human intervention is required. For example, during hardware servicing, cleaning, or recalibration of the system, manual control ensures that the user can manage the system without interference from the automated processes.

Adaptability in Extreme Weather Conditions: In cases of extreme weather, such as unusually high humidity due to heavy rainfall or other unusual atmospheric conditions, users can opt to override the automatic system to prevent unnecessary ventilation or to actively regulate the environment based on personal judgment.

User Preferences: Some users might prefer to maintain the flap in a particular position regardless of the changing humidity levels. The Manual Override Mode gives users the ability to manually control the flap position to suit their immediate comfort or needs without waiting for sensor-triggered actions.

Maintenance and Diagnostics: During periods of system maintenance or diagnostics, manual control allows engineers or technicians to test the hardware, run system checks, and make adjustments without interference from the automatic humidity regulation. The manual toggle provides a straightforward method for ensuring that the system's physical components are functioning as expected.

Ensured Control during Unforeseen Circumstances: In the event of sensor malfunction, electrical issues, or any form of unexpected disruption in the automatic functionality, the Manual Override Mode acts as a fail-safe mechanism, ensuring that the user can continue to operate the system effectively.

Operational Details: Activation and Deactivation: The capacitive touch module is used to toggle between the Automatic Mode and Manual Override Mode. The touch interface is responsive and requires minimal user interaction to switch modes, making it user-friendly

Predefined Positions: Once the Manual Override Mode is activated, the servo motor moves between specific open and closed positions that have been preconfigured based on user preferences. These positions are not influenced by the humidity levels, allowing for precise control of the ventilation flap's movement

By incorporating the Manual Override Mode, the system becomes significantly more versatile, accommodating the dynamic needs of users. This mode enhances the system's reliability and ensures that the user retains control over the environment in all situations, providing a balance of autonomy and flexibility.

The Manual Override Mode makes the Smart Home Ventilation System not only efficient but adaptable to various real-world situations, ensuring that users have complete control over their indoor environment, regardless of changing conditions or operational challenges.

3.3. Remote Control:

The Smart Home Ventilation System is equipped with remote control functionality through the Blynk IoT platform. This feature allows users to interact with the system from anywhere using their smartphones. By integrating Blynk, the system provides a seamless way to monitor and control the ventilation process without being physically present. Using the Blynk app, users can view live humidity readings in real-time, ensuring they are always aware of the indoor environmental conditions. The app also allows users to toggle between

automatic and manual modes with the touch of a button, offering flexibility and control based on personal preferences. In automatic mode, the system operates based on preset humidity levels, while in manual mode, users can override the system and control the servo motor directly via the app. This remote control feature adds a layer of convenience and makes the system adaptable to varying circumstances, such as when users are away from home or want to adjust the ventilation remotely. The integration of Blynk ensures that the system remains user-friendly and easily accessible, giving homeowners the ability to optimize their indoor air quality at any time, from anywhere.

3.4. Power Management:

A key aspect of the design is ensuring efficient power management for the servo motor while maintaining the protection and stability of the microcontroller. This is achieved by employing a carefully calibrated power management system, which regulates the voltage and current to prevent overloading or power fluctuations. By prioritizing power efficiency, the system guarantees reliable operation even under varying load conditions, ensuring consistent performance for the servo motor and microcontroller.

3.5. Scalability:

The modularity of the system design provides flexibility for future expansions. It supports the easy addition of new sensors, such as CO2, temperature, or air quality sensors, which can further enhance the system's functionality. Moreover, the design is built with cloud connectivity in mind, allowing for real-time data access and remote monitoring. This scalability ensures that the system can evolve to meet emerging needs, and it can seamlessly integrate with broader smart home ecosystems. The flexible architecture makes the system adaptable to future advancements in technology and user requirements.

3.6. Features:

The Smart Ventilation System integrates several thoughtful design elements and functional features to ensure optimal performance, user convenience, and adaptability. Below is a detailed overview of its key features:

• ***Real-Time Humidity Monitoring:***

Continuously monitors ambient humidity using a DHT11 sensor, ensuring that any deviation from the ideal moisture range is detected promptly and acted upon. • Autonomous Operation: Automatically regulates ventilation based on humidity thresholds without requiring user input, ensuring consistent air quality and energy savings.

• ***Manual Override Capability:***

Equipped with a capacitive touch switch that allows users to take direct control of the ventilation flap. Ideal for use during maintenance, troubleshooting, or when custom air circulation is desired.

• ***Servo Motor Actuation with Offset Handling:*** Controls a Servo motor with fine-tuned offset settings to ensure smooth and precise movement of the ventilation flap for effective air exchange.

• ***Dual-Mode Functionality:*** Seamless switching between automatic and manual modes ensures flexibility and user control in a variety of operating environments.

• ***Compact and Modular Design:*** Built with prototyping in mind, the system is easy to assemble, test, and upgrade. Components can be rearranged or scaled as needed for multiroom systems.

• ***Future-Ready Microcontroller:*** Utilizes the powerful ESP32, which includes built-in Wi-Fi and Bluetooth, allowing for future enhancements like mobile app control, remote monitoring, and cloud connectivity.

• ***Low Power Consumption:***

Designed for energy efficiency by utilizing low power components and duty cycling, making it suitable for continuous operation in home and office environments.

• ***Remote Control:***

One of the standout features of the Smart Home Ventilation System is its integration with the Blynk IoT platform, which enables remote control of the system via a smartphone app. With Blynk, users can monitor live humidity readings and easily switch between automatic and manual modes from anywhere, providing unparalleled convenience. The remote control functionality ensures that users can adjust the ventilation system even when they are not at home, ensuring continuous air quality control. This flexibility allows for a more personalized experience, as users can make adjustments based on changing conditions or preferences without needing to be physically present at the system's location.

3.7. Working Code: Below is the embedded C++ code for the ESP32. It handles sensor input, servo control, and touch sensor toggling with software debouncing**.**

```
#include <DHT.h>
#include <ESP32Servo.h>

#define BLYNK_TEMPLATE_ID "TMPL3LWAtZ7AB"
#define BLYNK_TEMPLATE_NAME "Smart Ventilation"
```

```
#define BLYNK_AUTH_TOKEN "Z4B71kh65EpL8ozAJYoMdDLQb4e5uaps"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

// WiFi and Blynk credentials
char auth[] = "Z4B71kh65EpL8ozAJYoMdDLQb4e5uaps";
char ssid[] = "OnePlus Nord CE4";
char pass[] = "smith2004";

#define DHTPIN 5
#define DHTTYPE DHT11
#define SERVO_PIN 14
#define TOUCH_PIN 27

DHT dht(DHTPIN, DHTTYPE);
Servo ventServo;

const int humidityThreshold = 85;
const int offset = 10;
bool overrideMode = false;
bool lastTouchState = false;
unsigned long lastDebounceTime = 0;
const unsigned long debounceDelay = 300;

int currentServoPos = 180;

// Blynk Switch Handler (V0): 0 = Auto, 1 = Manual
BLYNK_WRITE(V0) {
  int pinValue = param.asInt(); // 1 = Manual, 0 = Auto
  overrideMode = pinValue;

  Serial.println(overrideMode ? "Manual Override ON  [Blynk]" : "Auto Mode Enabled  [Blynk]");

  if (overrideMode) {
    if (currentServoPos > 90) {
      currentServoPos = 0 + offset;
```

```
    } else {
      currentServoPos = 180 + offset;
    }
    ventServo.write(currentServoPos);
    Serial.print("Manual Control (Blynk): Servo Position ");
    Serial.println(currentServoPos);
  }
}
void setup() {
  Serial.begin(115200);
  dht.begin();
  pinMode(22, OUTPUT);
  digitalWrite(22, HIGH);
  ventServo.attach(SERVO_PIN, 500, 2400);
  ventServo.write(currentServoPos);
  Blynk.begin(auth, ssid, pass);
  Serial.println("System Ready ");
}
void loop() {
  Blynk.run();  // Run Blynk tasks
  int touchValue = touchRead(TOUCH_PIN);
  Serial.print("Touch Value: ");
  Serial.println(touchValue);

  bool currentTouch = (touchValue < 30);
  unsigned long currentTime = millis();

  if (currentTouch && !lastTouchState && (currentTime - lastDebounceTime > debounceDelay)) {
    overrideMode = !overrideMode;
    Blynk.virtualWrite(V0, overrideMode);  // Sync switch in app
    Serial.println(overrideMode ? "Manual Override ON " : "Auto Mode Enabled ");

    if (overrideMode) {
      if (currentServoPos > 90) {
        currentServoPos = 0 + offset;
      } else {
        currentServoPos = 180 + offset;
```

```
    }

    ventServo.write(currentServoPos);
    Serial.print("Manual Control (Touch): Servo Position ");
    Serial.println(currentServoPos);
  }

  lastDebounceTime = currentTime;
}
lastTouchState = currentTouch;

if (!overrideMode) {
  float humidity = dht.readHumidity();

  if (isnan(humidity)) {
    Serial.println("Sensor Error");
  } else {
    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
    Blynk.virtualWrite(V1, humidity);
    if (humidity > humidityThreshold) {
      currentServoPos = 0 + offset;
      ventServo.write(currentServoPos);
      Serial.println("Vent OPEN ");
    } else {
      currentServoPos = 180 + offset;
      ventServo.write(currentServoPos);
      Serial.println("Vent CLOSED");
    }
  }
}

delay(500);
}
```

3.8. Block Diagram :

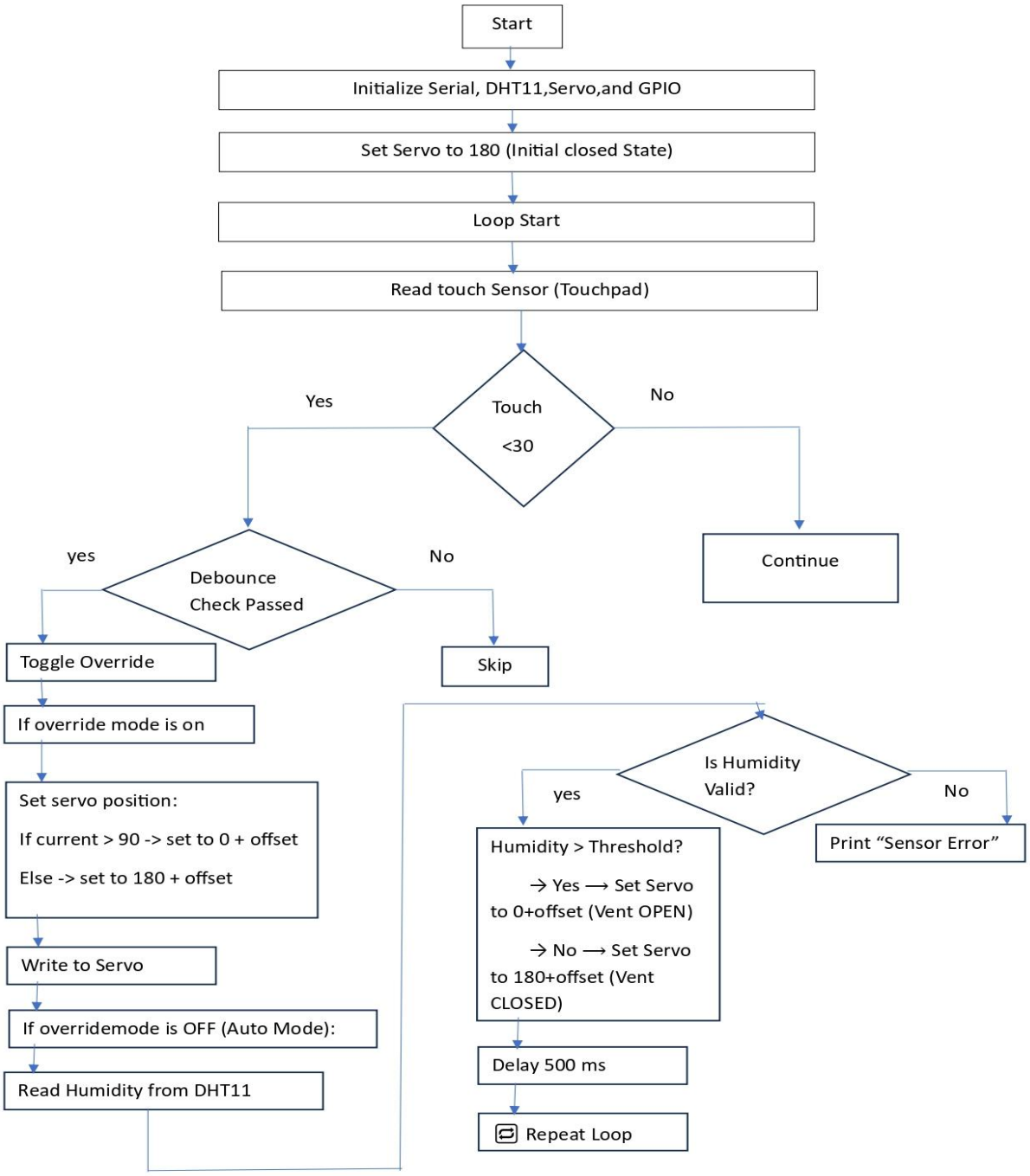

Figure 5

3.9. Circuit Diagram :

The wiring in the system plays a crucial role in ensuring that the various components of the circuit are properly connected and function as intended. It involves the integration of the ESP32's General Purpose Input/Output (GPIO) pins with the peripheral devices such as sensors, motors, and actuators. These connections are designed with attention to detail to ensure proper signal flow, and each device is connected to the corresponding GPIO pin on the ESP32 to facilitate seamless communication and control.

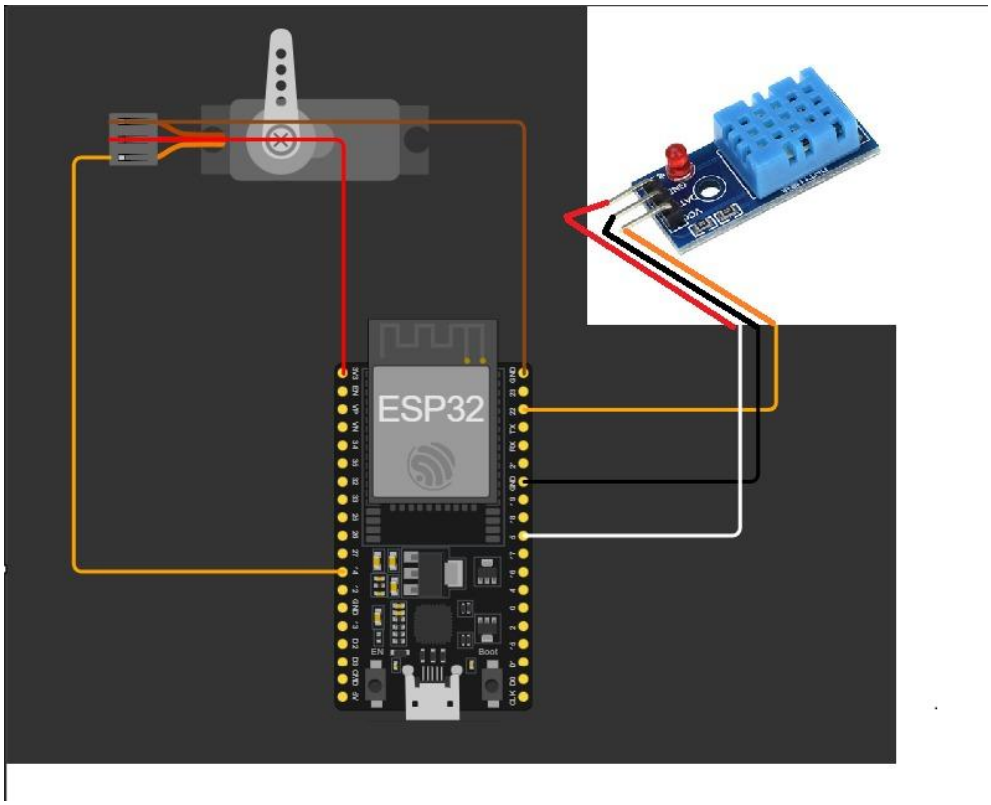

Figure 6

Signal integrity is one of the key factors in the wiring design. To prevent issues such as noise or voltage drops, resistors are used strategically in the circuit to regulate the current and protect sensitive components from potential damage. Additionally, for devices like motors, where sudden changes in current or voltage can cause harmful spikes, a flyback diode is employed. The flyback diode is essential for protecting the circuit from the back emf (electromotive force) generated by motors, particularly servos, when they suddenly stop or change direction. Without this protection, the voltage spike could damage the ESP32 or other sensitive components. Thus, the wiring not only facilitates the physical connections between the components, but also ensures the stability, durability, and longevity of the system by safeguarding it against potential electrical hazards.

Chapter 4

TESTING & RESULTS

4.1. Testing

Component Verification:

Each component (ESP32, DHT11, Servo motor) was individually tested to ensure correct functionality before integrating into complete circuit.

Power Supply Check: Voltage levels were monitored to ensure all components received proper and stable power, preventing malfunctions.

Sensor Accuracy: The DHT11 sensor was tested for consistent humidity readings by comparing its output under different environmental conditions.

Servo Motor Response: The servo's movement was checked for smooth rotation and correct angle control based on humidity levels.

ESP32 Upload Test: The microcontroller was tested to confirm successful code upload and serial communication with the PC.

Touch Override Functionality: The ESP32's touch pin was tested to verify that manual override correctly reset or disabled the servo operation.

Blynk Integration Testing: The system was connected to the Blynk IoT platform to verify remote monitoring and control features. The virtual switch was tested to toggle between automatic and manual override modes via mobile. The gauge widget was tested to ensure real-time humidity readings were accurately displayed on the Blynk dashboard.

Network Reliability: Wi-Fi connectivity was monitored for consistent Blynk communication. Reconnection routines were verified to handle occasional signal drops.

Mobile Control Responsiveness: Blynk app commands were tested for immediate effect on the hardware, ensuring minimal latency between mobile input and device response.

Field Testing: The complete system was tested in real indoor conditions with varying humidity levels to observe real-time adaptation and ventilation control.

User Interaction: The user interface via the Blynk app was found to be intuitive, allowing easy switching between control modes and live monitoring without technical difficulty.

Overall Integration: Final testing involved checking the coordination between all modules, ensuring automatic ventilation, manual override, and remote Blynk controls worked as expected under varying humidity conditions.

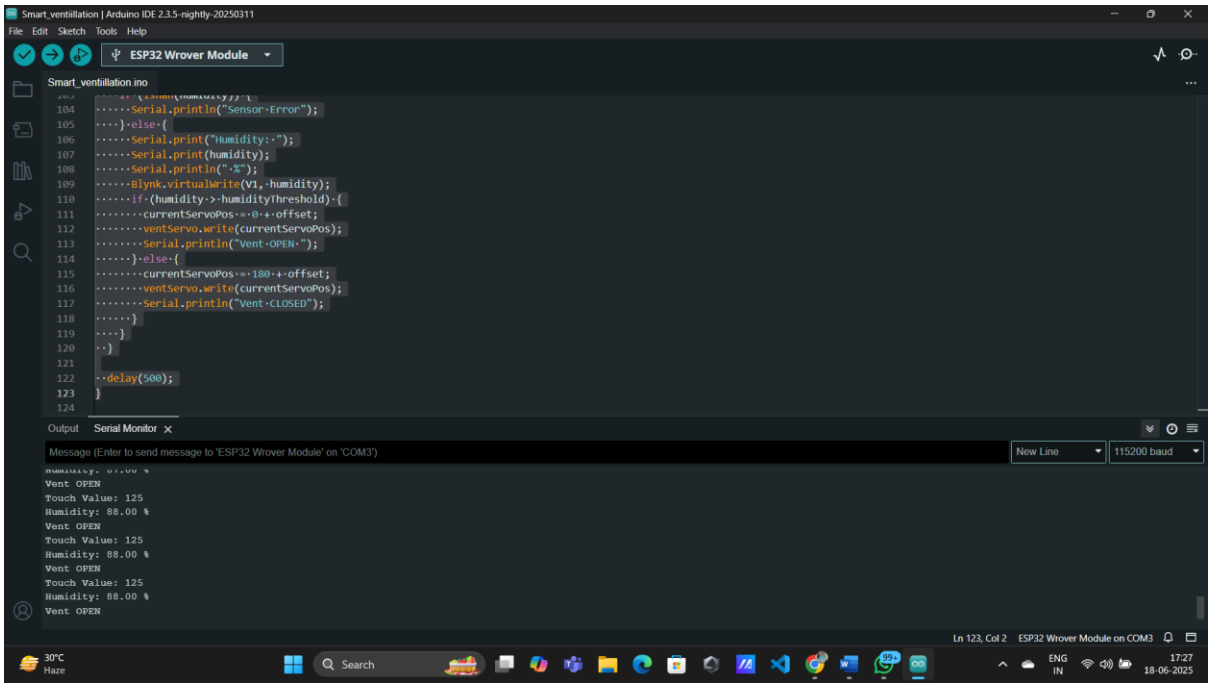

Figure 7 :Serial Monitor Output -Vent Open Operation

This output displays the humidity level detected by the DHT11 sensor crossing the defined threshold. As a result, the ESP32 sends a signal to activate the servo motor and open the vent. The message confirms that the system logic responds appropriately to elevated humidity levels, ensuring proper ventilation when required.

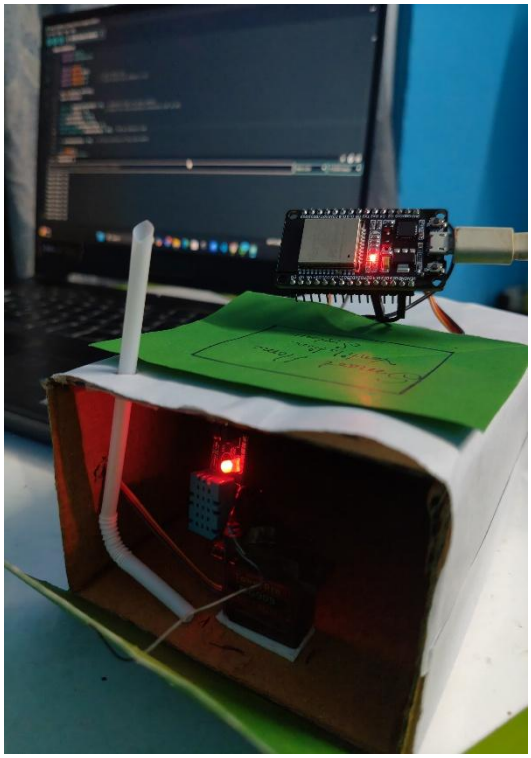

Figure 8 -Vent Open

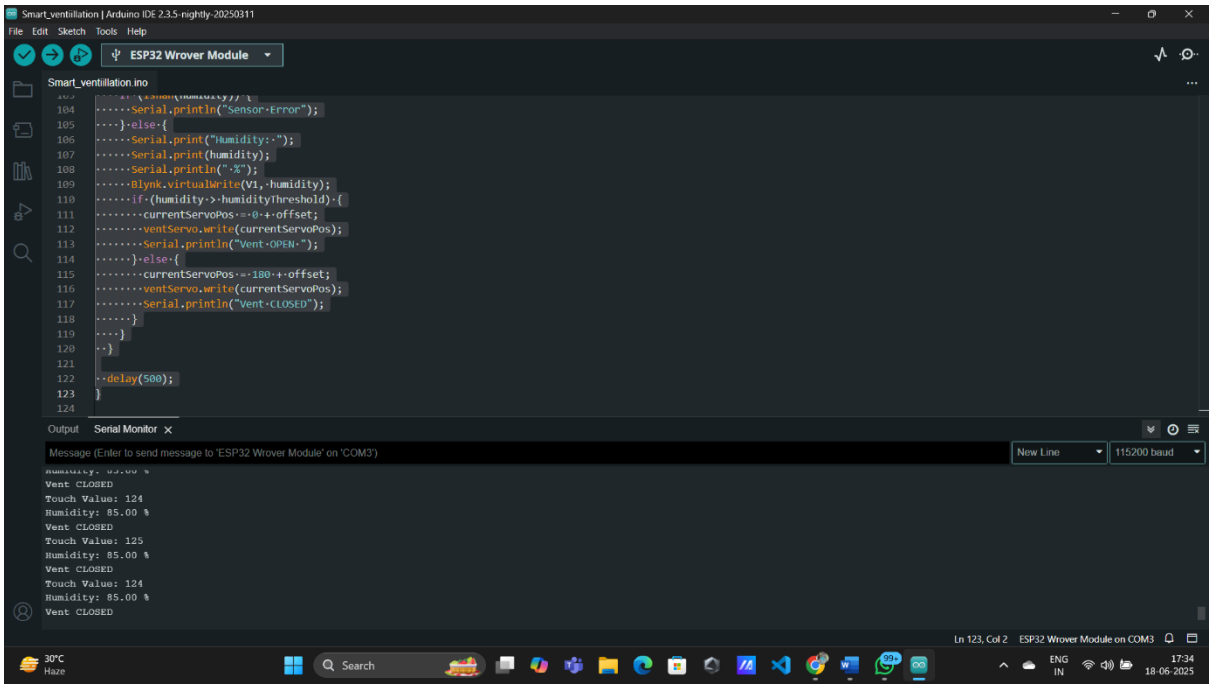

Figure 9 :Serial Monitor Output -Vent Closed Operation

This image captures the system response when humidity is below the preset threshold. The ESP32 keeps the servo motor in the default closed position, conserving energy and preventing unnecessary airflow. The serial output confirms that the system accurately maintains the vent's closed state under normal conditions.

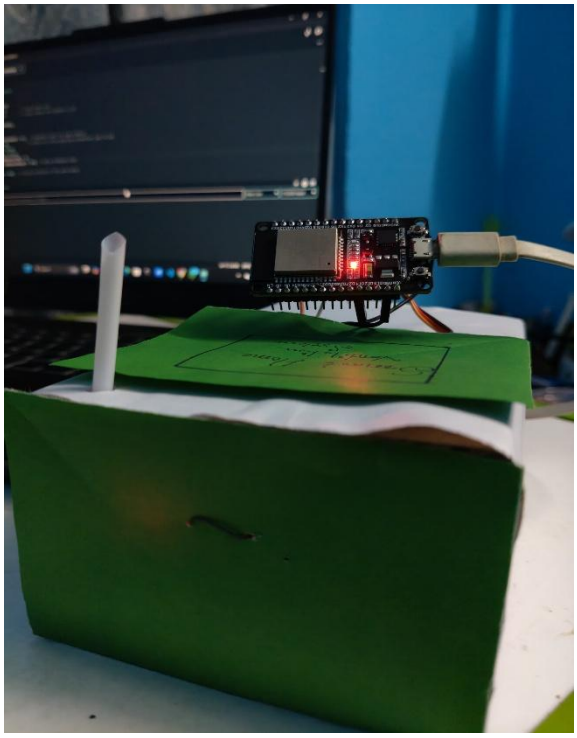

Figure 10 -Vent Closed

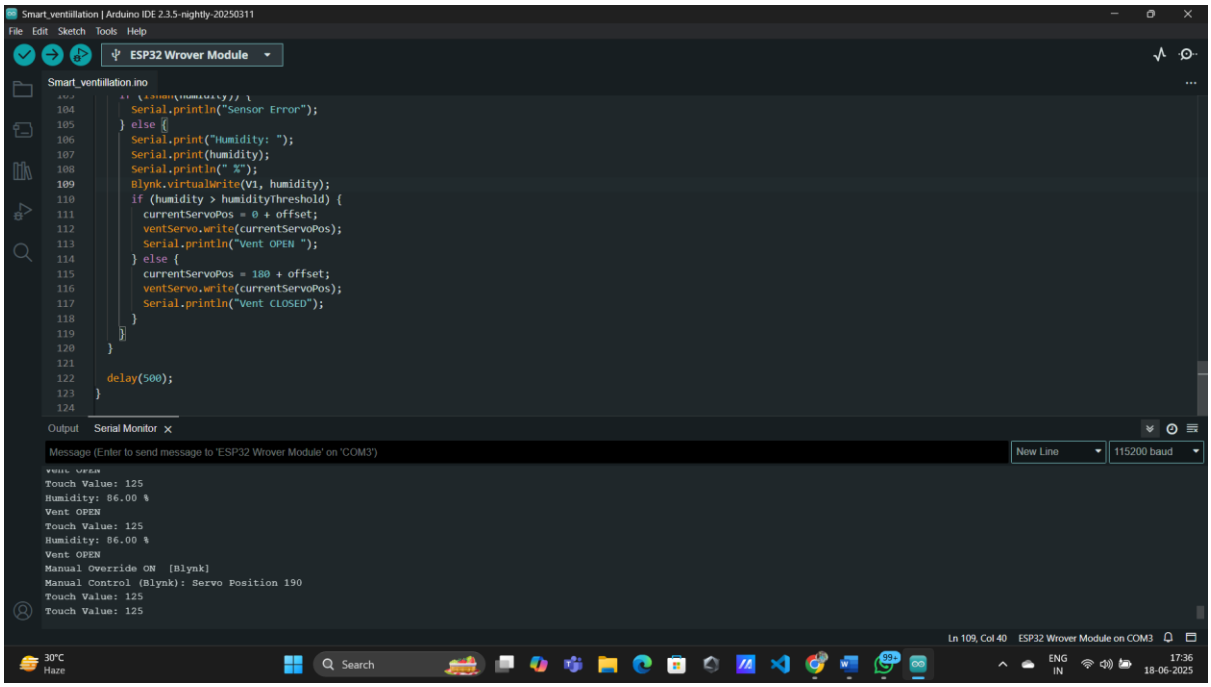

Figure 11 :Serial Monitor Output – Manual Override On

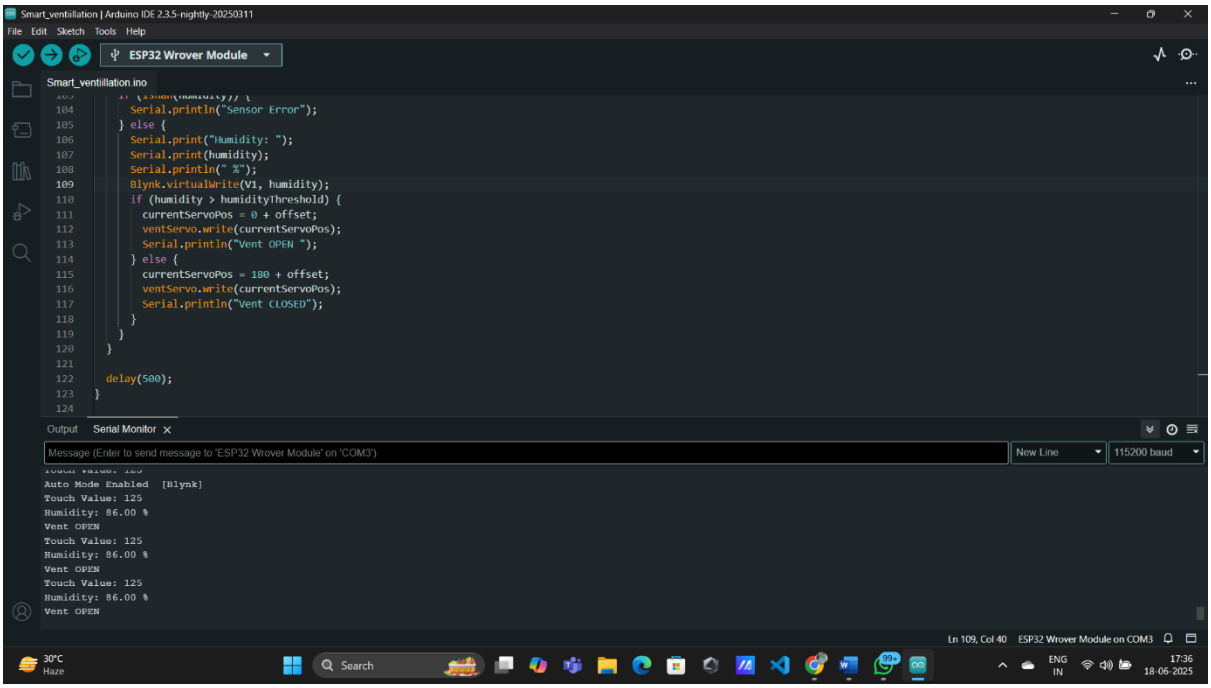

Figure 12 :Serial Monitor Output – Auto Mode Enabled(Manual Override Off)

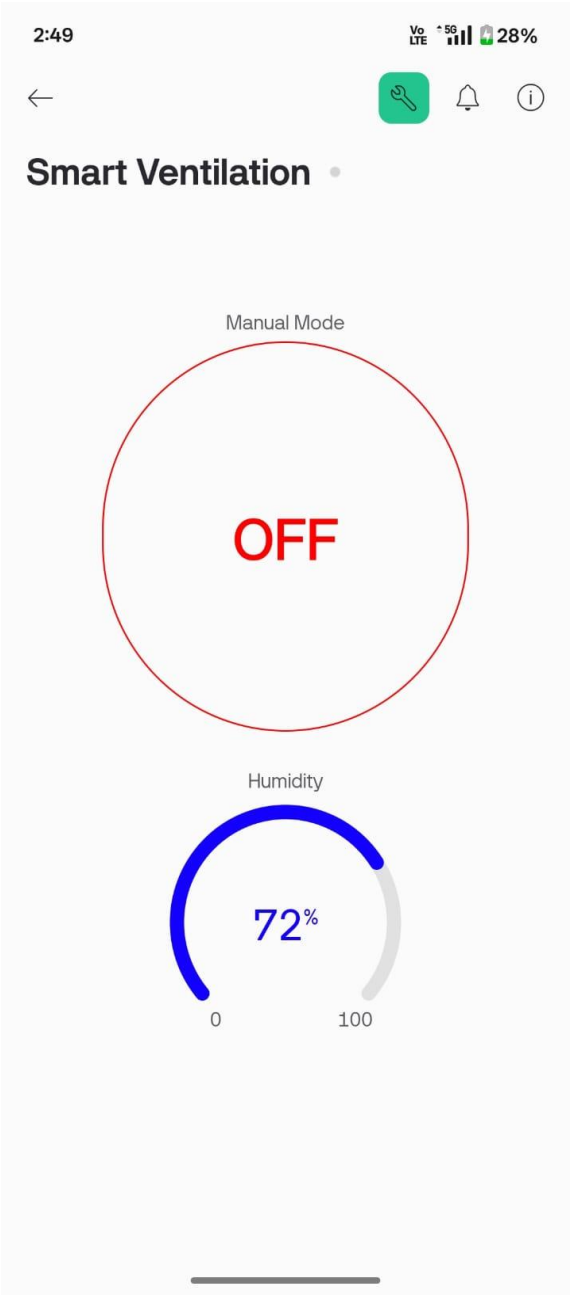

Figure 13 : Blynk Layout

4.2. Results :

TEST SCENARIO	HUMIDITY LEVEL	SERVO ANGLE	VENT CONDITION
Humidity > 85 (Auto Mode)	> 85	180	Vent Open
Humidity < 85 (Auto Mode)	<85	0	Vent Closed
Touch Press (Manual Mode): Initial Vent Closed	-	0→180	Vent Open
Touch Press (Manual Mode): Initial Vent Open	-	180→0	Vent Closed

TABLE 01

The Smart Home Ventilation System performed successfully under various testing conditions. When the DHT11 sensor detected humidity levels above the predefined threshold, the ESP32 microcontroller activated the servo motor, which in turn adjusted the ventilation flap to improve airflow. This automatic control ensured a responsive and energy-efficient ventilation process. During low humidity conditions, the system kept the flap closed, preserving indoor air quality without unnecessary operation. The override touch sensor also worked effectively, allowing manual control of the servo motor when needed. This added flexibility enhanced the system's usability and gave users direct control when desired. All components, including the ESP32, DHT11, and servo motor, communicated seamlessly, and the overall setup proved stable, consistent, and accurate. The response time was quick, with real-time data processing and action, showcasing the capability of ESP32 in handling sensor-driven applications. The results validate the practicality and reliability of using such a setup for smart home environments, providing comfort, automation, and control with minimal power consumption and cost.

Chapter 5

CONCLUSION

The Smart Home Ventilation System demonstrates a low-cost, reliable solution to manage indoor humidity dynamically. Its combination of real-time sensing, responsive actuation, and user-friendly manual control makes it suitable for practical deployment in homes, greenhouses, and small offices. The architecture is scalable, making it an ideal foundation for more complex smart environment systems.

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